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and Computer Engineering**

**Τμήμα Ηλεκτρολόγων Μηχανικών
και Μηχανικών Υπολογιστών**

Thesis - Διπλωματική Εργασία

“Talking Heads”: Phoneme Prediction from Muted Video

“Talking Heads”: Πρόβλεψη Φωνημάτων από Βουβό Βίντεο

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Summer 2026

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1 Summaries

1.1 Summary in English

Summary in English.

1.2 Περίληψη στα Ελληνικά

Περίληψη στα Ελληνικά.

2 Introduction

2.1 Statement of the Problem

What is the problem this work aims to solve? Why is it important? What may be some applications?

2.2 Available Methods

Briefly describe the methods that have been shown to (partially) solve this problem (or subproblems that may exist). Mention Section 3 (State of the Art) for more info.

2.3 Description of this Work

What did I do for this thesis? How did I attempt to solve the problem? About one page.

2.4 Structure of this Report

Describe the structure of this report in about half a page.

3 Background

Describe every necessary background element to understand this work. Tools, Definitions etc. Also define a notation (for the input data, the output data, ...) and stick to it for the rest of the Report.

4 State of the Art

A more detailed description of the state of the art I alluded to in the Introduction chapter.

5 Methodology

How was the problem solved? Detailed description of what I did. This is expected to be the lengthiest part of the Report.

6 Evaluation of the Methodology

6.1 Datasets

Which datasets were used for training in each of the experiments? This is the first mention of the datasets!!!

6.2 Evaluation Criteria

Define and describe the evaluation criteria (metrics) used in each experiment (loss curves, area under curve, accuracy, error rate, ...)

6.3 Experiments

Present a series of experiments, showcase and compare the results of each experiment to the state of the art and finally comment on each experiment. The following experiments are subject to change:

6.3.1 Experiment 1: Phoneme Prediction from Constrained Muted Video

What I've done so far.

6.3.2 Experiment 2: Phoneme Prediction from Unconstrained (In the Wild) Muted Video

6.3.3 Experiment 3: Phoneme Prediction using both visual and audio cues

7 Conclusions and Future Improvements

What have I understood from this project? What can I improve in the future?

8 References

- [1] Marshall Thomas, Edward Fish, and Richard Bowden. VALLR: Visual ASR Language Model for Lip Reading, 2026.